CO3-AT2-Case Study

CSA-0603-DESIGN OF ANALYSIS AND ALGORITHMS

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Title: Employee Database Search (Binary Search)

A corporate system maintains employee records in sorted order for efficient lookup. Analyze how Binary Search can improve retrieval speed. Identify requirements such as sorted structure and fast access. Apply divide and conquer techniques to explain the process. Analyze time complexity and justify the chosen approach.

Description of the Real-World Problem

In a corporate organization, thousands of employee records are maintained in a database. HR departments and management frequently need to search for employee information such as employee ID, name, department, or salary.

If employee records are stored in sorted order based on Employee ID, searching each record one by one using Linear Search can take more time when the database contains a large number of employees.

Therefore, an efficient searching algorithm is required to retrieve employee records quickly. Binary Search can significantly improve the search speed by repeatedly dividing the sorted employee records into smaller sections.

For example, a company with thousands of employee records can use Binary Search to quickly locate an employee using their Employee ID.

Problem Statement

Design an Employee Database Search System that searches employee records efficiently using the Binary Search Algorithm.

The system should:

- Accept multiple employee records.
- Maintain employee records in sorted order based on Employee ID.
- Search for a specific Employee ID efficiently.
- Reduce the number of comparisons.
- Handle large employee datasets efficiently.
- Analyze the best, average, and worst-case performance.
- Retrieve the employee record when the Employee ID is found.

Algorithm Used

Binary Search Algorithm

Binary Search is an efficient Divide and Conquer searching algorithm. It works only when the data is arranged in sorted order.

It works in three main steps:

1. Find the middle element of the sorted employee records.
2. Compare the middle Employee ID with the required Employee ID.
3. If the required ID is smaller, search the left half; if it is larger, search the right half.
4. Repeat the process until the employee is found or the search range becomes empty. **Justification for Choosing the Algorithm (Why Binary Search?)** Binary Search is selected because:

- It is much faster than Linear Search for large sorted datasets.
- Its average and worst-case time complexity is O(log n).
- It follows the Divide and Conquer technique.
- It reduces the search area by half after every comparison.
- It is suitable for searching large employee databases.
- It requires fewer comparisons compared with Linear Search.
- It provides efficient retrieval when employee records are sorted.

For example, searching among 1,000 employee records using Binary Search requires only a small number of comparisons compared with checking records one by one.

Divide and Conquer Principle

Binary Search follows the Divide and Conquer approach.

Divide

Find the middle employee record and divide the search range into two halves.

Conquer

Compare the target Employee ID with the middle Employee ID and continue searching only in the appropriate half.

Repeat

Continue dividing the selected half until the employee is found or no records remain.

Combine

No separate combining step is required because Binary Search directly returns the position of the required employee.

Working Example (Manual Execution)

Suppose the Employee IDs are:

[101, 105, 110, 115, 120, 125, 130]

Search Employee ID = 125

Step 1

Original

101 105 110 115 120 125 130

Middle Element = 115

Compare:

125 > 115

So, search the **right side**.

Right Side

120 125 130

Step 2

Middle Element = 125

Compare:

125 == 125

Employee ID 125 is found.

Final Result

Employee ID 125 found at index 5

Search Process

101 105 110 |115| 120 125 130

↓

Right Side

↓

120 |125| 130

↓

Element Found

Binary Search Performance

Binary Search repeatedly divides the sorted employee records into two halves. **Good Case**

- Employee ID is found at the middle position.
- Time Complexity: **O(1)**

Average and Worst Case

- Search space is divided by half in every step.
- Time Complexity: **O(log n)**

Complexity Analysis

Case	**Time Complexity**
Best Case	O(1)
Average Case	O(log n)
Worst Case	O(log n)

Explanation

- **Best Case – O(1):** The target Employee ID is found at the middle position during the first comparison.
- **Average Case – O(log n):** The search range is repeatedly reduced by half.
- **Worst Case – O(log n):** The algorithm continues dividing the search range until the element is found or the range becomes empty.

Space Complexity

O(1) for an iterative Binary Search implementation.

The algorithm uses only a few variables such as low, high, and mid, so no additional data structure is required.

Applications

Binary Search can be used in:

- Employee Database Management Systems
- Human Resource Management Systems
- Payroll Systems
- Banking Systems
- Library Management Systems
- Student Record Management
- Database Search Systems
- Inventory Management Systems
- Customer Record Management

Advantages

- Very fast for large sorted datasets.
- Average and worst-case time complexity is **O(log n)**.
- Reduces the search space by half in every iteration.
- Requires fewer comparisons than Linear Search.
- Simple and efficient for sorted employee records.
- Uses **O (1)** extra space in iterative implementation.
- Suitable for large databases with fast random access.

Limitations

- Employee records must be sorted before searching.
- It is not suitable for unsorted data without first sorting it.
- Efficient random access to records is required.
- Maintaining sorted records can require additional processing when records are frequently inserted or deleted.

Comparison with Other Searching Algorithms

Algorithm	Best	Average	Worst	Requirement
Linear Search	O(1)	O(n)	O(n)	No sorting required
Binary Search	O(1)	O(log n)	O(log n)	Sorted data

Sample Python Program

```python
# ========================================
# Employee Database Search using Binary Search
# ========================================

def binary_search(employee_ids, key):
    low = 0
    high = len(employee_ids) - 1
    while low <= high:
        mid = (low + high) // 2
        if employee_ids[mid] == key:
            return mid
        elif employee_ids[mid] < key:
            low = mid + 1
        else:
            high = mid - 1
    return -1
employee_ids = list(map(int, input(
    "Enter sorted Employee IDs: "
).split()))
key = int(input("Enter Employee ID to search: "))
position = binary_search(employee_ids, key)
if position != -1:
    print("Employee ID found at index:", position)
else:
    print("Employee ID not found.")
```

```
Enter sorted Employee IDs: 101 105 110 115 120 125 130
Enter Employee ID to search: 125
Employee ID found at index: 5


...Program finished with exit code 0
Press ENTER to exit console.
```

Sample Input

Enter sorted Employee IDs: 101 105 110 115 120 125 130

Enter Employee ID to search: 125

Sample Output

Employee ID found at index: 5

Conclusion

Binary Search is an efficient algorithm for searching employee records maintained in sorted order. By applying the **Divide and Conquer** technique, it reduces the search space by half after every comparison. This results in **O(log n)** average and worst-case time complexity, making it much faster than Linear Search for large sorted employee databases. Therefore,

Binary Search is a suitable approach for efficient employee record retrieval in corporate database systems.